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Atomic data of intermediate autoionzing Rydberg series $\mathbf{nf[K]_J}$ $\mathbf{(n = 4, 5)}$, $\mathbf{nd[K]_J}$ $\mathbf{(n = 5, 6)}$, $\mathbf{np[K]_J}$ $\mathbf{(n = 6, 7)}$ and $\mathbf{ns[K]_J}$ $\mathbf{(n = 7, 8)}$ of neutral argon atom in the multiconfiguration Dirac-Hartree-Fock framework*

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Abstract. We report the results of an accurate computation of the energy levels, oscillator strengths f_{ij}, the radiative transition rates A_{ij}, the Landé g-factor, the magnetic dipole and the electric quadrupole hyperfine constants of the intermediate Rydberg series $n\mathrm{f}[K]_J$ $(n = 4, 5)$, $n\mathrm{d}[K]_J$ $(n = 5, 6)$, $n\mathrm{p}[K]_J$ $(n = 6, 7)$ and $n\mathrm{s}[K]_J$ $(n = 7, 8)$ relative to the ground state $3\mathrm{p}^6\ {}^1\mathrm{S}_0$ for neutral argon atom spectra based on the multiconfiguration Dirac-Hartree-Fock (MCDHF) approach. In this approach, the quantum electrodynamics (QED) effects with transverse photon Breit interaction and vacuum photon polarization self-energy corrections are included. Moreover, the perturbation due to the core-polarization and electron correlation effects are taken into account. We also reported the oscillator strengths and transition rates in Coulomb and Babushkin gauges for the $3\mathrm{p}^5(^2\mathrm{P})3\mathrm{d}$-$3\mathrm{p}^5(^2\mathrm{P})4\mathrm{f}$, $3\mathrm{p}^5(^2\mathrm{P})3\mathrm{d}$-$3\mathrm{p}^5(^2\mathrm{P})5\mathrm{f}$, $3\mathrm{p}^5(^2\mathrm{P})4\mathrm{d}$-$3\mathrm{p}^5(^2\mathrm{P})4\mathrm{f}$, $3\mathrm{p}^5(^2\mathrm{P})4\mathrm{d}$-$3\mathrm{p}^5(^2\mathrm{P})5\mathrm{f}$, $3\mathrm{p}^5(^2\mathrm{P})5\mathrm{d}$-$3\mathrm{p}^5(^2\mathrm{P})4\mathrm{f}$, $3\mathrm{p}^5(^2\mathrm{P})5\mathrm{d}$-$3\mathrm{p}^5(^2\mathrm{P})5\mathrm{f}$, $3\mathrm{p}^5(^2\mathrm{P})6\mathrm{d}$-$3\mathrm{p}^5(^2\mathrm{P})4\mathrm{f}$ and $3\mathrm{p}^5(^2\mathrm{P})6\mathrm{d}$-$3\mathrm{p}^5(^2\mathrm{P})5\mathrm{f}$ transition arrays as well as the $3\mathrm{p}^5(^2\mathrm{P})5\mathrm{d}$-$3\mathrm{p}^5(^2\mathrm{P})6\mathrm{p}$, $3\mathrm{p}^5(^2\mathrm{P})7\mathrm{s}$-$3\mathrm{p}^5(^2\mathrm{P})6\mathrm{p}$, $3\mathrm{p}^5(^2\mathrm{P})7\mathrm{s}$-$3\mathrm{p}^5(^2\mathrm{P})7\mathrm{p}$ and $3\mathrm{p}^5(^2\mathrm{P})8\mathrm{s}$-$3\mathrm{p}^5(^2\mathrm{P})7\mathrm{p}$ transition arrays. Insofar as experimental or theoretical values by other authors exist, comparison is made with those values; it turns out that, for most of the levels, a rather satisfying agreement with these experimental and theoretical results is obtained, thus confirming the reliability of our data.

1 Prelude

Over the years, the spectroscopic data such as the energy levels, the oscillator strengths f_{ij}, the radiative transition rates A_{ij}, the Landé g-factor, the magnetic dipole and the electric quadrupole hyperfine constants of argon atoms have been of significant interest to various applications such as Ar-Kr-F_2 laser system [1], X-ray lasers [2–4], plasma physics, atmospheric sciences and astrophysics [5–8]. This is one of the several reasons why these atoms have been, over the years the subject of numerous experimental and theoretical studies. Specifically, these atoms possess two distinctive ionization limits called $I_{3/2}$ and $I_{1/2}$, corresponding to two odd-parity states ${}^2\mathrm{P}_{3/2}$ and ${}^2\mathrm{P}_{1/2}$ of the $3\mathrm{p}^5$ ionic core, respectively. These limits of ionization are $I_{3/2} = 127\,109.9\,\mathrm{cm}^{-1}$ $\pm 0.1\,\mathrm{cm}^{-1}$ and $I_{1/2} = 128\,541.8\,\mathrm{cm}^{-1}$ $\pm 0.1\,\mathrm{cm}^{-1}$ and give birth to a double set of singly excited Rydberg level series converging to each of these two ionization limits [9]. These series exhibit remarkable features. Below the first limit of ionization, these series overlap and interact with each other resulting in perturbations, whereas autoionization effects take place above the first limit of ionization. Therefore, their experimental data are difficult to be analyzed, because the classical method of detection is insufficient to detect the electronic transitions from the ground state to the states of these series.

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The spectroscopic notations of the fine-structure levels issues from the $3p^5nl'$ electronic configuration of argon are usually designated in the jlk-coupling scheme, commonly called Racah coupling [10]. In this coupling, the total angular momentum j_c, of the $3p^5$ core couples with the orbital angular momentum l of the Rydberg electron to form the intermediate quantum number k. The latter couples with the spin, s, of the Rydberg electron to give the total-angular-momentum quantum number J. The fine-structure levels belonging to the terms $3p^5nl'$ are usually designated by $3p^5nl[k]_J$ and $3p^5n'l'[k]_J$ depending on the state of the core $^2P_{3/2}$ and $^2P_{1/2}$, respectively.

The energy positions and the line shapes of the Rydberg series attached to the two distinct threshold of ionization of the argon atom have been extensively studied experimentally [9,11–23] as well as theoretically [10,12,24–27]. However, most of these experimental and theoretical data are limited to few transitions among the terms belonging to odd-parity (ns, ns$'$, nd, and nd$'$) [15–17,21–23] or even-parity of (np, np$'$, nf, and nf$'$) Rydberg series [12–21]. Heretofore, the most accurate energy positions are those reported by Yoshino [9]. These measurements have been realized by absorption of VUV light from the ground state and restricted for the sharp ns$'$, $J = 1$ autoionizing resonances to levels below the first limit of ionization. Minnhagen [11] successfully measured the energy of the levels issued from the configuration $3p^5nl$ using the grating spectrograph.

The energy positions and line shapes of the levels belonging to the even-parity $n^5P_{1/2}14p'$ and $9f'$ ($n = 3, 4$, and 5) autoionizing resonance were reported by Petrov and co-workers [12] by using one photon excitation from lower-lying intermediate levels. Also, they calculated the corresponding line shapes. Disparity between their experimental and theoretical results is observed. Evidently, more experimental results are required in order to test their calculations. Peters and co-workers [13] investigated both experimentally and theoretically several np$'$ autoionizing resonances of the argon atom by UV excitation.

Yin-Yu Lee and co-workers [14] have investigated the autoionizing Rydberg series np$'[1/2]_0$ ($n = 11$–57), np$'[3/2]_2$ ($n = 11$–65), np$'[3/2]_1$ ($n = 11$–31) and nf$'$ ($n = 9$–74) by stepwise excitations with lasers and synchrotron radiation. Based on optogalvanique laser spectroscopy, the $3p^5ns'[1/2]_{0,1}$ ($11 \leq n \leq 34$) autoionizing resonances of argon have been investigated by Landais and co-workers [16]. They used two-step excitation from the $3p^54s[3/2]_{1,2}$ populated in a discharge. Moreover, Klar and co-workers [17] have carried out an extremely high-resolution experiment using a two-photon excitation from a metastable beam, they have observed for the first time the ns$'$ $J = 0$ resonances, but only for $n = 18$ and 25. Pellarin and co-workers [18] have employed collinear laser spectroscopy techniques jointly with a field ionization detection method in order to measure the even-parity spectra of two hundred and forty levels belonging to the even-parity Rydberg series $3p^5np$ and $3p^5nf$. Also, they used the Multichannel Quantum Defect Theorem (MQDT) in order to calculate the energy of the whole series.

New experimental energy data of the autoionizing $3p^5np'$ (n ranging from 11 to 39) and $3p^5nf'$ (n ranging from 9 to 18) were reported by Chun-Yan lee and co-workers [19].

Recently, Charlotte Froese Fischer and co-workers [25] computed the energy of the levels belonging to the series $ns[k]_J$ ($4 \leq n \leq 5$), $3d[k]_J$ and $4p[k]_J$ of neutral argon atom. More recently, Salah and Hassouneh [28] have calculated the energy of the levels belonging to the intermediate Rydberg series $ns[k]_J$ ($4 \leq n \leq 6$), $n\mathrm{d}[k]_J$ ($3 \leq n \leq 4$), $np[k]_J$ ($4 \leq n \leq 5$) relative to the ground state $3p^6\ ^1S_0$ for neutral argon atom spectra. These values are obtained in the framework of the multiconfiguration Dirac-Hartree-Fock (MCDHF) approach. In this approach, Breit interaction, leading quantum electrodynamics (QED) effects and self-energy correction are taken into account. Moreover, their data are in good agreement with those measured experimentally by Minnhagen [11] and much better than those predicted by Charlotte Froese Fischer and co-workers [25].

The oscillator strengths of the transitions in the discrete and continuum regions of noble gases represent an important step towards various applications in plasma and astrophysics. The knowledge of the oscillator strengths of atomic system can be used in order to calculate the optical properties of atoms, transition rates for absorption and emission, as the lifetimes of the excited states which might be difficult to observe. However, various experimental techniques [29–35] and theoretical models [36–39] have been employed in order to measure experimentally or to calculate theoretically the oscillator strengths of neutral argon atoms. Among these experimental works, we quote the measurement of optical oscillator strength by Baig and co-workers [29]. They used a two-step laser excitation optogalvanique detection in order to measure the optical oscillator strength of the dominant autoionizing Rydberg series $3p^5(^2P_{1/2})\,nd'[3/2]_2$ and $nd'[5/2]_{2,3}$ ($n = 13$–52) in argon. Chan and co-workers [30] used a dipole (e, e) spectrometer to measure the absolute photoabsorption oscillator strength for argon in the energy range 16–500 eV. The self-absorption of the resonance radiation method was employed to measure the optical oscillator strengths of the noble-gas resonance lines by Lintenberg [31], Gibson and Risely [32]. The absolute oscillator strengths for the discrete and autoionizing resonances were reported by Wu and co-workers by means of electron-energy-loss spectroscopy [33]. Westerveld and coworkers [34] measured experimentally the oscillator strengths using the self-absorption of resonance radiation for fourteen resonance transitions of neon and argon. The angle-resolved-electron-energy-loss spectrometer is used by Ji and coworkers [35] in order to determine the absolute generalized oscillator strengths of the 4s, 4s$'$, 4p + 4p$'$ excitation of argon. In a different manner, based on the MCDHF method, Salah and Hassouneh [28] calculated the oscillator strength of the levels belonging to the intermediate Rydberg series $ns[k]_J$ ($4 \leq n \leq 6$), $n\mathrm{d}[k]_J$ ($3 \leq n \leq 4$), $np[k]_J$ ($4 \leq n \leq 5$) relative to the ground state $3p^6\ ^1S_0$ for neutral argon atoms as well as the for 3d, 4s-4p, 4p-3d, 4p-5s, 5s, 4d, 5s-5p, 5p-6s transition arrays.

The transition probabilities A_{ij} and lifetimes for some levels belonging to the Rydberg autoionizing series were measured experimentally or predicted theoretically by various authors. Chornay and co-workers used the electron-photon delayed coincidence to measure the radiative lifetimes of the $3p^5 4s$ excited state [40]. Das and Karmakar measured the lifetime of some levels belonging to the $3p^5 nd$ $(n = 3, 4, 5, 6, 7)$ by employing high-frequency deflection techniques [41]. Aymar and co-workers [42] predicted the theoretical transition probabilities and lifetimes of few levels for neon, argon krypton and xenon.

Lately, argon theoretical energies and transition probabilities of $J = 1$ odd and $J = 2$ even states have been calculated by Savukov [27]. In his calculation, a combined configuration-interaction and many-body perturbation theory were used. Furthermore, the transition rates for the levels issue from the intermediate Rydberg series $n\mathrm{s}[k]_J$ $(4 \le n \le 6)$, $n\mathrm{d}[k]_J$ $(3 \le n \le 4)$, $n\mathrm{p}[k]_J$ $(4 \le n \le 5)$ relative to the ground state $3p^6\ ^1S_0$ for neutral argon atom spectra as well as the for 3d, 4s-4p, 4p-3d, 4p-5s, 5s, 4d, 5s-5p, 5p-6s transition arrays have been calculated by Salah and Hassouneh [28] using the MCDHF method.

The experimental and theoretical values of the Landé g-factor occupy an interesting place in atomic spectroscopy. These values are of great interest in a variety of physics applications. They could be used to determine the magnetic dipole A_{MD} and the electric quadrupole B_{EQ} hyperfine constants in cross level experiment [43–45]. Hitherto, the experimental values for Landé g-factor of neutral argon atoms are suffering from shortcoming and are far from being complete. Experimental measurements [15,45,46] and theoretical predictions [47] of the Landé g-factor of neutral argon atoms were undertaken by Salah [15], Chenevier and Moskowitcz [46] and Aymar and Schweighofer [47] as well as NIST [48]. Salah measured the Landé g-factor for the levels $6\mathrm{d}[1/2]_1$, $7\mathrm{d}[1/2]_1$, $8\mathrm{d}[1/2]_1$ and $9\mathrm{d}[1/2]_1$ by using optogalvanique atomic laser spectroscopy. The uncertainty of his results is estimated to be 0.02. On the other hand, Chenevier and Moskowitcz [46] measured experimentally the Landé g-factor for the levels $3\mathrm{d}[1/2]_1$ using optically detected magnetic resonance technique, while the theoretical value of this level, among others, is computed by Aymar and Schweighofe [47]. Abu Safia measured and theoretically calculated the Landé g-factor for the level $4\mathrm{p}[1/2]_1$, $4\mathrm{p}'[1/2]_1$, $4\mathrm{p}'[3/2]_2$, $4\mathrm{p}'[3/2]_1$, $4\mathrm{p}[3/2]_2$, $4\mathrm{p}[3/2]_1$ and $4\mathrm{p}[5/2]_2$ [45,49]. In his measurements, Abu Safia used laser optical pumping and magnetic resonance technique.

Moreover, Salah and Hassouneh [28] calculated the relativistic Landé g-factor, the magnetic dipole and the electric quadrupole hyperfine constants of the intermediate Rydberg series $n\mathrm{s}[k]_J$ $(4 \le n \le 6)$, $n\mathrm{d}[k]_J$ $(3 \le n \le 4)$, $n\mathrm{p}[k]_J$ $(4 \le n \le 5)$ relative to the ground state $3p^6\ ^1S_0$ for neutral argon atom spectra. Nevertheless, the Landé g-factor, the magnetic dipole A_{MD} and the electric quadrupole B_{EQ} hyperfine constants of the neutral argon atom are scarce.

After all, most of the experimental and theoretical data of the spectroscopic parameters of the neutral argon atom available in the literature are restricted to energy levels and limited to a few transitions among the terms belonging to the highly excited states of the Rydberg series.

In spite of a wealth of dedicated experimental results and theoretical prediction of spcectroscopic parameters for highly excited levels belonging to the even- and odd-parity Rydberg series, the knowledge of spectroscopic data of many low-lying levels of these series remains unsystematic and relatively scant and far from being complete. The scarceness of these data restricts our ability to deduce assuredly the properties of many cosmic plasma and hence many of the fundamental issues in astrophysics [50]. In order to fill this lack, to provide new reliable atomic data for low-lying levels belonging to the autoionized Rydberg series and to complete our previous results [28], this paper has been conceived.

In this paper, we report sufficiently accurate theoretical data of the energy, the oscillator strengths f_{ij}, the radiative transition rates A_{ij}, the Landé g-factor, the magnetic dipole and the electric quadrupole hyperfine constants for many levels belonging to the intermediate Rydberg series $n\mathrm{f}[k]_J$ $(n = 4, 5)$, $n\mathrm{d}[k]_J$ $(n = 5, 6)$, $n\mathrm{p}[k]_J$ $(n = 6, 7)$ and $n\mathrm{s}[k]_J$ $(n = 7, 8)$. Since the transitions from the ground state to the levels 4f and 5f are not allowed either by parity or electric dipole selection rules, we calculated all the oscillator strengths and transition rates in emission for $3p^5(^2P)3d$-$3p^5(^2P)4f$, $3p^5(^2P)3d$-$3p^5(^2P)5f$, $3p^5(^2P)4d$-$3p^5(^2P)4f$, $3p^5(^2P)4d$-$3p^5(^2P)5f$, $3p^5(^2P)5d$-$3p^5(^2P)4f$, $3p^5(^2P)5d$-$3p^5(^2P)5f$, $3p^5(^2P)6d$-$3p^5(^2P)4f$ and $3p^5(^2P)6d$-$3p^5(^2P)5f$ transition arrays. Similarly, for the level 5p, we calculated all the transitions $3p^5(^2P)5d$-$3p^5(^2P)6p$, $3p^5(^2P)7s$-$3p^5(^2P)6p$, $3p^5(^2P)7s$-$3p^5(^2P)7p$ and $3p^5(^2P)8s$-$3p^5(^2P)7p$ transition arrays.

These spectroscopic parameters are calculated in the framework of MCDHF method. The correction of energy due to the dominant part of the radiative quantum electrodynamics effects included transverse photon Breit interaction and vacuum photon polarization, self-energy, the core-polarization and electron correlation effects are included.

2 Brief theoretical procedure

In the multiconfiguration Dirac-Haretree-Fock (MCDHF) approximation, the atomic state wave function (ASF) is represented as a linear combination of symmetry-adapted configuration state wave functions (CSFs)

$$\Psi\left(\Pi, J, M_J\right) = \sum_{r}^{N_{CSF}} c_r \psi\left(\gamma_r \Pi, J, M_J\right), \tag{1}$$

which are eigenfunctions of the parity operator Π, and the total-angular-momentum operators J_z and J^2.

Here γ_r represents the additional quantum numbers to define each ASF or CSFs only, c_r being the configuration mixing coefficients. These coefficients are obtained via the diagonalization of Dirac-Coulomb Hamiltonian in atomic unit:

$$H_{DC}=\sum_{i=1}^{N}\left[c\widehat{\alpha}\cdot\hat{P}_i+(\widehat{\beta}_i-\hat{I})c^2-\frac{Z}{r_i}\hat{I}\right]+\sum_{i<j}\left[\frac{1}{r_{ij}}+V_B(i,j)\right], \quad (2)$$

with

$$V_B(i,j)=-\frac{1}{2r_{ij}}\left[\widehat{\alpha}_i\cdot\widehat{\alpha}_j+\frac{1}{r_{ij}^2}\left(\widehat{\alpha}_i\cdot\vec{r}_{ij}\right)\left(\widehat{\alpha}_j\cdot\vec{r}_{ij}\right)\right], \quad (3)$$

where $V_B(i,j)$ is the frequency-independent Breit part of the electron-electron interaction, the indices $i,j=1,2,3\ldots,N$ numerate the electrons, $c=137$ being the speed of light in atomic units, the first term on the right-hand side of eq. (2) is the one-particle Dirac Hamiltonian while the second term represents the electrostatic interaction of the electrons with each other, and r_{ij} represents the relative position of the electrons, $\frac{Z}{r_i}$ is the monopole scalar potential due to the electron-nucleus interaction, $\hat{P}=-i\nabla$ is the electron momentum operator, $\widehat{\alpha}=\left(\begin{smallmatrix}0&\widehat{\sigma}\\ \widehat{\sigma}&\hat{0}\end{smallmatrix}\right)$ is a four-matrix known as Dirac operator and $\widehat{\sigma}$ represents the Pauli matrices, and $\widehat{\beta}=\left(\begin{smallmatrix}\hat{I}&0\\ 0&-\hat{I}\end{smallmatrix}\right)$ is a four-matrix known as the Dirac basis where $\hat{I}$ is the unit matrix.

The dominant part of the radiative QED effects with vacuum photon polarization and self-energy corrections are included.

The standard relativistic spin-orbitals,

$$\psi_{nkm}\left(\theta,\phi,\sigma\right)=\begin{pmatrix}\varphi_{nkm}\\ \widetilde{\varphi}_{nkm}\end{pmatrix}=\frac{1}{r}\begin{pmatrix}iP_{nk}(r)\chi_{km}\left(\theta,\phi,\sigma\right)\\ Q_{nk}(r)\chi_{-k,m}\left(\theta,\phi,\sigma\right)\end{pmatrix}, \quad (4)$$

were employed in order to build a N-electron basis of configuration state functions (CSFs) in which the Hamiltonian matrix which can be obtained from eq. (2), to be later considered.

In eq. (4), $i=\sqrt{-1}$, n is the principle quantum number, m is the z-component of the total angular momentum $\vec{J}$, k is the relativistic angular-momentum quantum number commonly called Dirac quantum number

$$k=(-1)^{j+l+\frac{1}{2}}\left(j+\frac{1}{2}\right), \quad (5)$$

$P_{nk}(r)$ and $Q_{nk}(r)$ are the radial functions usually called the large and small components [51,52], and χ_{km} are spinor spherical harmonics [53]. These functions behave under rotation in space like the behavior of the spherical harmonics $Y_{l,m_l}(\theta,\phi,)$ in the non-relativistic case, whenever the orbital angular momentum $\vec{l}$ is replaced by the one electron total angular momentum $\vec{J}=\vec{l}+\vec{s}$. However, the angular part χ_{km} is a linear combination of spherical harmonics $Y_l^{m-m_s}$ and the spin function S which is a spinor of rank 2:

$$\chi_{km}\left(\theta,\phi,\sigma\right)=\sum_{m_s}Y_l^{m-m_s}S^{m_s}\left\langle l\frac{1}{2}m-m_s\middle|l\frac{1}{2}jm_j\right\rangle. \quad (6)$$

Under these assumptions, it is conceivable to treat all angular- and spin-dependent parts analytically.

3 Computation of the relativistic radiative transition rates A_{fi} and the absorption oscillator strength f_{if}

3.1 The relativistic radiative transition rates A_{fi}

The relativistic radiative transition rates A_{if} per second corresponding to the transition from the initial state i to the final state f for an electron outside a closed shell [52,54] is

$$A_{fi}=2\alpha\omega_{if}\frac{2j_i+1}{2L+1}\begin{pmatrix}j_f&L&j_i\\ 1/2&0&-1/2\end{pmatrix}^2|\overline{M}_{if}|^2, \quad (7)$$

with the matrix elements $\overline{M}_{if}$ for a relativistic radiative transition of a single electron multipole operator of order L [54]. These matrix elements are either the integral for electron multipole transitions or the integral for the electron magnetic multipole transitions [52].

These radiative transitions are defined as

$$M_{if} = \begin{cases} M_{if}^{(e)} + GM_{if}^{(1)}, & \text{for electric multipole transitions,} \\ M_{if}^{(m)}, & \text{for magnetic multipole transitions,} \end{cases} \tag{8}$$

where

$$M_{if}^{(m)} = (i)^{l+1} \frac{2L+1}{\sqrt{L(L+1)}} (k_f + k_i)\, I_L^{+}(\omega_{if}), \tag{9}$$

$$M_{if}^{(e)} = (i)^L \left[\sqrt{\frac{L}{L+1}} [(k_i - k_f) I_{L+1}^{+}(\omega_{if}) + (L+1) I_{L+1}^{-}(\omega_{if})] - \sqrt{\frac{L+1}{L}} [(k_i - k_f) I_{L-1}^{+}(\omega_{if}) - L I_{L-1}^{-}] \right], \tag{10}$$

$$M_{if}^{(1)} = (i)^L \left[(k_i - k_f) I_{L+1}^{+}(\omega_{if}) + (L+1) I_{L+1}^{-}(\omega_{if}) + \left[(k_i - k_f)\, I_{L-1}^{+}(\omega_{if})\, L I_{L-1}^{-} \right] + (2L+1)\, J_L \right], \tag{11}$$

with

$$I_L^{\pm} = \int_0^{\infty} j_l \left(\frac{\omega_{if}}{c} r \right) \left(P_{n_i k_i}(r) Q_{n_f k_f}(r) \pm Q_{n_i k_i}(r) P_{n_f k_f}(r) \right) \mathrm{d}r, \tag{12}$$

$$J_L = \int_0^{\infty} j_l \left(\frac{\omega_{if}}{c} r \right) \left(P_{n_i k_i}(r) P_{n_f k_f}(r) + Q_{n_i k_i}(r) Q_{n_f k_f}(r) \right) \mathrm{d}r, \tag{13}$$

where j_l is the spherical Bessel function and G in eq. (8) is the gauge parameter. It takes the value 0 in Coulomb gauge [55,56] and $\sqrt{\frac{L+1}{L}}$ in Babushkin gauge [57–59]. In the non-relativistic limit, $G = 0$ gives the velocity form of radiation matrix elements, while $G = \sqrt{\frac{L+1}{L}}$ gives the length form.

3.2 The relativistic absorption oscillator strength f_{if}

The dimensionless absorption oscillator strength f_{if} for a transition from the initial level i to the final level f are calculated in the single multipole approximation. In this approximation, the interference between different multipoles transitions will not be taken into account, whereas transition rates corresponding to arbitrary multipoles will be taken into account. Under these conditions, the line strength f_{if} for the transition from initial atomic state function Ψ_i to final-state function Ψ_f induced by a multipole radiation field operator $\hat{D}_m^{(L)}$ of order L is

$$f_{if} = \frac{\pi c}{(2L+1)\omega_{if}^2} |\langle \Psi_i | \hat{D}_m^{(L)} | \Psi_f \rangle|^2. \tag{14}$$

The last equation can be written in terms of the reduced matrix elements as

$$f_{if} = \frac{\pi c}{(2L+1)\omega_{if}^2} \sum_{\mu\nu} b_{\mu f} b_{i\nu} \langle n_i k_i || \hat{D}_m^{(L)} || n_f k_f \rangle, \tag{15}$$

where the reduced matrix elements $\langle n_i k_i || \hat{D}_m^{(L)} || n_f k_f \rangle$ are given by

$$\langle n_i k_i || \hat{D}_m^{(L)} || n_f k_f \rangle = \left(\frac{(2j_i+1)\omega_{if}}{\pi c} \right)^2 (-1)^{j_i - \frac{1}{2}} \begin{pmatrix} j_i & L & j_f \\ 1/2 & 0 & -1/2 \end{pmatrix} |\overline{M}_{if}|. \tag{16}$$

4 Evaluation of the Landé g-factor

4.1 Prelude

The relativistic Hamiltonian $\hat{H}$ that represents the interaction of an atom with a uniform magnetic field $\vec{B}$ is given by

$$\hat{H} = \frac{1}{2} \sum_i e \widehat{\alpha}_i \cdot \left(\vec{B} \times \vec{r}_i \right), \tag{17}$$

where the summation is taken over all atomic i-th electron, e is the electron charge. For simplicity, we shall drop the index i in the following equations. In terms of the scalar product of magnetic dipole moment spherical tensor of rank 1, $\mu_q^{(1)}$, eq. (17) becomes

$$\hat{H} = \frac{1}{2}\sum \mu_q^{(1)} \cdot \vec{B}, \tag{18}$$

with

$$\sum \mu_q^{(1)} = -\sum ie\sqrt{\frac{8\pi}{3}} r\hat{\alpha} \cdot Y_{lq}^{(0)}(\vec{r}), \tag{19}$$

where $Y_{lq}^{(0)}(\vec{r})$ represent the spherical-harmonic vector.

The Landé g-factor g_J is defined by the magnetic dipole moment operator $\hat{\mu}$ of an atom in the state $|JM_J\rangle$ as

$$\hat{\mu} = -g_J \mu_B \hat{J}, \tag{20}$$

where μ_B is the Bohr magneton.

Since matrix elements of the Hamiltonian given by eq. (20) can be calculated in the state $|JM_J\rangle$, the matrix elements of the Hamiltonian $\hat{H}$ are

$$\langle \hat{H} \rangle = g_J \mu_B \langle \hat{J} \cdot \vec{B} \rangle, \tag{21}$$

then the Landé g-factor g_J can be written by means of Wigner-Eckart theorem [60] as

$$g_J = \frac{1}{2\mu_B} \frac{\langle J \| \sum \mu_q^{(1)} \| J \rangle}{\sqrt{J(J+1)}} \,. \tag{22}$$

On the other hand, the well-known spin factor g_s can be expressed in the form [61]

$$g_s = 2\left(1 + \frac{\alpha}{2\pi} - 0.3828\left(\frac{\alpha}{\pi}\right)^2 + \cdots\right), \tag{23}$$

where $\alpha = \frac{1}{137}$ is the fine-structure constant.

The correction to the electron g_s factor leads to a correction to the relativistic interaction Hamiltonian by an amount

$$\Delta\hat{H} = 0.001160 \mu_B \hat{\beta}\hat{\alpha} \cdot \vec{B}, \tag{24}$$

and this leads to a correction to Landé g-factor g_J by an amount

$$\Delta g_J = 0.001160 \frac{\langle J \| \sum \mu_q^{(1)} \| J \rangle}{\sqrt{J(J+1)}} \,. \tag{25}$$

4.2 Computation of the radial matrix elements

The relativistic eigenfunction for a state denoted in Dirac notation by $|\Gamma_J \Pi J M_J\rangle$ can be expanded in jj-coupled configuration state functions which are eigenfunctions of J^2, J_z and the parity operator Π in the MCDHF as

$$|\Gamma_J \Pi J M_J\rangle = \sum_r C_r |\gamma_J \Pi J M_J\rangle, \tag{26}$$

where Γ_J and γ_J represent the configuration and any other quantum number required to specify a state, and C_r are the expansion coefficients. On the other hand, the configuration state functions are sums of product of four-components spin-orbital function named Dirac wave function. The spin spherical harmonics χ_{km} defined in eq. (6) can be written in the LSJ-coupling as

$$\chi_{km} = \sum_q C\left(l\frac{1}{2}j; m - qqm\right) Y_{lm-q}(\vec{r}). \begin{cases} \begin{pmatrix} 1 \\ 0 \end{pmatrix}, & q = \dfrac{1}{2}\,, \\[2ex] \begin{pmatrix} 0 \\ 1 \end{pmatrix}, & q = -\dfrac{1}{2}\,. \end{cases} \tag{27}$$

Moreover, an angular recoupling computer program [62] is used in order to reduce the matrix elements $\langle J\| \sum \mu_q^{(1)} \|J\rangle$ and $\langle J\| \sum \Delta\mu_q^{(1)} \|J\rangle$ to terms involving single-particle orbitals only,

$$\left\langle J \left\| \sum \mu_q^{(1)} \right\| J \right\rangle = \sum_{a,b} D_{ab}^{(1)}(rs) \left\langle n_a k_a \left\| \mu^{(1)} \right\| n_b k_b \right\rangle, \tag{28}$$

$$\left\langle J \left\| \sum \Delta\mu_q^{(1)} \right\| J \right\rangle = \sum_{a,b} D_{ab}^{(1)}(rs) \left\langle n_a k_a \left\| \Delta\mu^{(1)} \right\| n_b k_b \right\rangle, \tag{29}$$

where

$$D_{ab}^{(1)} \equiv -\sum (-1)^{\frac{1}{2}} e \sqrt{\frac{8\pi}{3}} r\widehat{\alpha} \cdot Y_{lq}^{(0)}(\vec{r}) \delta_{b\acute{b}} \sqrt{2j_a+1}\sqrt{2j_b+1} \cdot (-1)^{a+b+j_b+1} \cdot \left\{ \begin{matrix} b & a & j_a \\ 1 & j_b & \acute{a} \end{matrix} \right\}, \tag{30}$$

and the $\{\}$ denotes the $6j$-symbol.

However, the single-particle matrix elements can be reduced into angular factors and radial integral:

$$\left\langle n_a k_a \left\| \mu^{(1)} \right\| n_b k_b \right\rangle = -\alpha\,(k_a + k_b) \left\langle n_a k_a \left\| C^{(1)} \right\| n_b k_b \right\rangle [r]_{n_a k_a n_b k_b}, \tag{31}$$

$$\left\langle n_a k_a \left\| \Delta\mu^{(1)} \right\| n_b k_b \right\rangle = (k_a + k_b - 1) \left\langle n_a k_a \left\| C^{(1)} \right\| n_b k_b \right\rangle [r^{(0)}]_{n_a k_a n_b k_b}, \tag{32}$$

where

$$\left\langle k_a \left\| C^{(1)} \right\| k_b \right\rangle \begin{cases} (-1)^{j_b - 1/2} \sqrt{2j_b+1} \begin{pmatrix} j_a & J_b & 1 \\ 1/2 & -1/2 & 0 \end{pmatrix}, & \text{if } l_a + 1 + l_b \text{ is even} \\ 0, & \text{if } l_a + 1 + l_b \text{ is odd} \end{cases} \tag{33}$$

$$[r]_{n_a k_a n_b k_b} = \int_0^\infty r\left(P_{n_a k_a}(r) Q_{n_b k_b}(r) + Q_{n_a k_a}(r) P_{n_b k_b}(r)\right) \mathrm{d}r, \tag{34}$$

$$\left[r^{(0)}\right]_{n_a k_a n_b k_b} = \int_0^\infty r^{(0)} \left(P_{n_a k_a}(r) P_{n_b k_b}(r) + Q_{n_a k_a}(r) Q_{n_b k_b}(r)\right) \mathrm{d}r. \tag{35}$$

4.3 Computation of the hyperfine interaction

In relativistic framework, the hyperfine Hamiltonian is

$$\hat{H}_{hfs} = \sum_i T_n^{(k)}(j) \cdot T_e^{(k)}(i), \tag{36}$$

where $T_n^{(k)}$ and $T_e^{(k)}$ are the spherical tensor operator of rank k, representing the nuclear and electron angular momentum, respectively, j and i designate the various protons and electrons, respectively.

The mean value of $\hat{H}_{hfs}$ in a fine-structure state $|JIFM_F\rangle$ is given to first order as

$$\langle \hat{H}_{hfs} \rangle = \sum_i \delta_{M_F M_{\acute{F}}} \delta_{F\acute{F}} (-1)^{F+J+I} \left\{ \begin{matrix} F & J & I \\ k & I & \acute{J} \end{matrix} \right\} \langle I||T_n^{(k)}||I\rangle \cdot \langle J||T_e^{(k)}||\acute{J}\rangle, \tag{37}$$

where I is the nuclear spin and $\vec{F} = \vec{I} + \vec{J}$ is the total angular momentum of the atom.

For the magnetic dipole case, $k = 1$, while the in the electric quadrupole case $k = 2$.

For the sake of simplicity, we express the nuclear dipole moment in nuclear magneton μ_n units, the nuclear dipole moment μ_I can be written as

$$\mu_I \mu_n = \langle IM_I = I|T_{n0}^{(k)}|IM_I = I\rangle = \begin{pmatrix} I & k=1 & I \\ -I & 0 & I \end{pmatrix} \langle I||T_n^{(1)}||I\rangle. \tag{38}$$

On the other hand, the operator $T_q^{(1)}$ is given by [63,64]

$$T_q^{(1)} = \sum t_q^{(1)} = -\sum_j ie\sqrt{\frac{8\pi}{3}} \frac{1}{r_j^2} \vec{\alpha}_j \cdot Y_{1q}^{(0)}(\hat{r}_j) \tag{39}$$

where e is the absolute magnitude of the electron charge and j denotes the j-th electron in the atom.

4.3.1 Calculation of the magnetic dipole hyperfine constant A_{MD}

The hyperfine constant A_{MD} is related to the nuclear magnetic moment by

$$A_{MD} = \mu_n \frac{\mu_I}{I} \langle J||T^{(1)}||J\rangle (J(J+1)(2J+1))^{-1/2}. \tag{40}$$

By means of eq. (37), the hyperfine energy splitting E_{MD} due to the magnetic dipole is

$$E_{MD} = \frac{A_{MD}}{2}(F(F+1) - I(I+1) - J(J+1)). \tag{41}$$

4.3.2 Calculation of the electric quadrupole constant B_{EQ}

The electric quadrupole constant B_{EQ} is related to the nuclear quadrupole moment Q. The latter is given by

$$eQ = 2\langle I, M_I = I|Q_0^{(2)}|I, M_I = I\rangle = \begin{pmatrix} I & k=2 & I \\ -I & 0 & I \end{pmatrix} 2\langle I||Q_0^{(2)}||I\rangle. \tag{42}$$

Since e is the charge and Q is the operator for the proton, then eQ represent the charge reparation in the nucleus.

On the other hand, the operator $T_q^{(2)}$ is given by [65] as

$$T_q^{(2)} = \sum t_q^{(2)} = \sum_j \frac{-e}{r_j^3} \sqrt{\frac{\pi}{(2k+1)}} Y_{qk}. \tag{43}$$

Here, the summation is carried out over all electrons of the atom.

The electric quadrupole hyperfine constant becomes

$$B_{EQ} = 2eQ \left[\frac{2J(2J-1)}{(2J+1)(2J+2)(2J+3)}\right]^{1/2} \langle J||T^{(2)}||J\rangle. \tag{44}$$

By means of eq. (44), the hyperfine energy splitting E_{EQ} due to the electric quadrupole is

$$E_{EQ} = \frac{B_{EQ}}{8} \frac{(3C(C+1) - 4J(J+1)I(I+1))}{J(2J-1)I(2I-1)}, \tag{45}$$

$$C = F(F+1) - I(I+1) - J(J+1). \tag{46}$$

5 Numerical results and discussions

5.1 Method of calculation

The above equations are plugged in a power full General Relativistic Structure Package code called GRASP2K [66]. This package is based on the fully relativistic (four-component) multiconfiguration Dirac-Hartree-Fock (MCDHF) method. To perform the calculation, the core $1s^2 2s^2 2p^6$ was assumed to be inactive. Under this assumption, an expansion of configuration state functions (CSFs) derived from an orbital set was used. Moreover, the restrictive active space method was used to generate CSFs. These CSFs are linear combinations of Slater determinants of atomic orbitals, and are simultaneous eigenfunctions of the atomic electronic total-angular-momentum operator, J, and the atomic parity operator. A linear combination of these CSFs is employed in order to generate an approximate atomic state functions (ASFs).

The CSFs were generated by excitation of orbitals occurring in the reference set of configurations to a set of orbitals. These orbitals are determined by optimization of an energy expression for an LS term.

A stepwise increasing of the orbital set was used in order to monitor the convergence of our numerical results. In the MCDHF approach, orbitals are determined by optimization of an energy expression for an LS term. As soon as the radial functions are determined, relativistic configuration interaction (RCI) calculations were carried out to determine CSF expansion coefficients. These coefficients were determined by diagonalizing the Hamiltonian matrix that included the frequency-dependent Breit interaction, vacuum polarization and self-energy correction.

The radial functions and expansion coefficients were enhanced to self-consistency using the multiconfiguration self-consistent field (MCSCF) procedure [67]. The corresponding eigenvalues and eigenvectors are determined by means of the iterative Davidson method [68]. After that, we used these eigenvectors in order to calculate the required allowed transition data between levels of odd and even parities. The allowed selection rules for electric dipole transitions in the jk coupling scheme are: $\Delta J = 0, \pm 1$, $\Delta j = 0$, $\Delta k = 0, \pm 1$ and $\Delta l = \pm 1$. Nevertheless, these rules are not entirely respected, since transitions with change of the ionic core as well as transitions with $\Delta l = 3$ are also observed [18]. These transitions are allowed because of the small mixing of the initial $4s[3/2]_2$ level with some levels of the $3p^5 3d$ configuration. A multiconfigurational computation fulfilled in the frame of central-field approximation [69] has shown that the $4s[3/2]_2$ is mixed with the $3d[3/2]_2$ and $3d'[3/2]_2$ levels but is not coupled to the $3d[5/2]_2$ and $3d'[5/2]_2$ levels. Under these conditions, the electric dipole selection rules $\Delta J = \pm 1$, $\Delta k = 0, \pm 1$ remain valid. Moreover, the electric dipole transition rule requires a parity change between the initial and final state.

For the computation of the energy levels of neutral argon atoms, a list of CSFS was obtained based on the notion of excitations from subshells to an active set of orbitals, *i.e.* the active set method. More precisely, for each specified parity Π and J symmetry (J^{Π}), the list of CSFs is generated, by taking into account the excitation from a given single reference configuration to a set of definite relativistic orbitals. By keeping all orbital subshells active, a permitted single excitation from $1s^2 2s^2 3s^2 3p^5 (^2P^\circ)$ reference configuration, to each one of nl relativistic orbitals, is considered. In our study, the principal quantum number n is in the range $3 \leq n \leq 8$ and l corresponds to all s, p, d and f orbitals under consideration. By taking the stability of the code into account, the calculations were done by separating each optimization stage corresponding to a given n and l for the outer shell orbital from another different optimization stage corresponding to n' and l' outer shell orbital. After the stage of relativistic state configuration function RSCF calculations, the RCI calculations stage is carried out. In the latter stage, both Breit interaction, quantum electrodynamics (QED) effects and self-energy correction were taken into account. Moreover, the expansions for the RCI calculations were obtained by allowing all single excitations to the $n \leq 8$ basis orbitals from the reference configurations, with keeping all subshells orbitals active.

Since many transitions of interest are found between the low-lying levels in the neutral argon atom, whose electronic configuration is [Ne] $3s^2 3p^6$, and the only parent of the Rydberg electron is $3p^5\ ^2P$, the core-polarization and electron correlation effects are taken into account. In this approximation, the ASFs are expanded over even $3s^2 3p^5\ n'l'$ configuration states and one set of orbitals is optimized. For the levels under consideration, only 4f, 5f, 5d, 6d, 6p and 7p were treated as valence ones and varied in the calculations which were fulfilled for $3 \leq n \leq 8$. For $n = 8$, only 8s orbitals were considered. To first-order correction by perturbation theory, the ASFs is written as a linear combination of all CSFs that are obtained via single replacement from the occupied orbital of the reference configuration to the virtual orbitals. For these virtual orbitals, we generated five layers containing the orbitals s, p, d, f and g. The latter orbitals were produced in a restrained configuration space. In this space, only a certain number of electron-pair correlations were counted in the MCDHF. Afterward, a series of CI calculations including different electron correlation effects are carried out in order to select the significant ones.

The hyperfine-structure constants and the Landé g-factors are calculated to first-order hyperfine interaction in the frame of the MCDHF method. Hence, the distortions of the electron shell by nuclear moments are not taken into account in the calculations [70,71]. The selected wave functions $|J, M_J\rangle$ used in MCDHF consist of all possible jj-coupled states that come up from the $3p^5 3d$ electronic configurations. By means of these wave functions, we built the eigenstates $|I, J, F, M_F\rangle$ of the hyperfine Hamiltonian. Here I and J being the total angular momentum of the nucleus and the electron state, respectively, and $\vec{F} = \vec{I} + \vec{J}$.

More details of the computational approach can be found in [72–74].

5.2 Computation of energy levels, oscillator strengths and transition rates in the neutral argon atom

We calculated the energy of the levels belonging to the intermediate autoionized Rydberg series $n\mathrm{f}[K]_J$ ($n = 4, 5$), $n\mathrm{d}[K]_J$ ($n = 5, 6$), $n\mathrm{p}[K]_J$ ($n = 6, 7$) and $n\mathrm{s}[K]_J$ ($n = 7, 8$). The energies of these levels with symmetry J^{Π} are predicted by diagonalizing the matrix of the Hamiltonian given by eq. (2) in the CSFs basis. The relativistic corrections to the energy levels due to Breit frequency, photon vacuum polarization, QED effect, self-energy correction, core-polarization and electron correlation effects are included. The obtained predicted results are illustrated in table 1 of the Supplementary Material (SM). In order to judge the accuracy of our predicted results, we compare them with those observed experimentally with uncertainty of $0.05\,\mathrm{cm}^{-1}$ by Minnhagen [11]. This comparison shows that our results are in good agreement with those reported by Minnhagen [11]. The differences between our results and Minnhagen ones are due to the calculation of the central potential for radial orbitals and recoupling schemes of angular parts. However, a part of this disparity can be attributed to the estimation of the ground-state energy which is slightly less than the corresponding experimental one. Also these differences might due to the degeneracy of the ^{1}D and ^{3}D terms of the p^5nd configuration, the relativistic corrections induced a strong admixture of terms in certain levels. In order to guarantee the reliability of our dissection, the resulting admixture terms were reproduced from the MCDHF wave functions. These wave functions were formed by a transformation from the jj to LSJ coupling. Since the LSJ coupling admixtures terms of the same angular momentum J, therefore, the coupling between levels are considered.

Unlike the odd-parity series ns and nd, the even-parity series np and nf are not well known. The odd-parity series can be reached by optical excitation of Ar from the ground state $3p^6\ ^1S_0$, however the even-parity series are optically forbidden by single-photon excitation so that they can be achieved only via the excited odd-parity Rydberg states. Because the atoms of intermediate excited state are difficult to prepare, the investigation of the even Rydberg series is a challenging task. For that, we have calculated the energy position of the levels belonging to the series $n\mathrm{f}[K]_J$ ($n = 4, 5$). During the calculation, more attention was paid in the interpretation of the energy levels belonging to these series. We noticed from table 1 of SM that the levels of each series overlap which, in turn, leads to a perturbation of the energy of these overlapping levels. The energy interval between two adjacent levels (the main level and the associated one) reduces towards the higher levels. The perturbation due to the interaction between the series $4\mathrm{f}[K]_J$ and $6\mathrm{P}[K]_J$ as well as between $5\mathrm{f}[K]_J$ and $7\mathrm{p}[K]_J$ were taken into account. Moreover, the $3p^5 4f[5/2]_3\ ^3D_3$ level is strongly coupled by spin-orbit interaction with the $3p^5 4f[5/2]_3\ ^3F_3$ levels. Similarly, the $4f[5/2]_2\ ^3F_2$ level is strongly coupled with the $4f[5/2]_2\ ^1D_2$ level. Indeed, the same spin-orbit coupling takes place in the series $5\mathrm{f}[K]_J$. In the latter series, the levels $^3D_3$ coupled with $^3F_3$ and $^3F_2$ coupled with $^1D_2$. On the other hand, the transitions from the ground state to the levels belonging to these series do not obey the electric dipole transition rules while they fulfill the electric quadrupole transition rules. In our calculations, stepwise excitations were considered in order to attain the levels of these series by satisfying the electric dipole selection rules.

In the $5\mathrm{d}[K]_J$ autoionized series, the $^3F_3$ level is coupled with the $^1F_3$ level and the $^3P_2$ level is coupled with the $^3D_2$ level by pure spin-orbit interaction. This interaction is responsible of the strong coupling between the levels belonging to the series $6\mathrm{d}[K]_J$, where $^3F_3$ is coupled with $^1F_3$, $^3P_2$ is coupled with $^3D_2$ and $^1P_1$ is coupled with $^3D_1$. Similarly, in the $7\mathrm{s}[K]_J$ and $8\mathrm{s}[K]$ series, the $^3P_1$ level is coupled with $^1P_1$.

In addition to spin-orbit interaction, the interaction between series is also responsible of the disparity between our results of the energy levels belonging to the series $7\mathrm{s}[K]_J$, $8\mathrm{s}[K]_J$, $6\mathrm{d}[K]_J$ and $7\mathrm{d}[K]_J$ and those measured by Minnhagen. Since the interaction between two series arises when the levels of one series have the same parity and total angular momentum J as the levels in the other series, the $7s[3/2]_2$ level is mixed with $6d[3/2]_2$ and $6d'[3/2]_2$, but it is not coupled with the $6d[5/2]_2$ and $6d'[5/2]_2$ levels. Although, the $8s[3/2]_2$ level is mixed with $7d[3/2]_2$ and $7d'[3/2]_2$, but it is not coupled with the $7d[5/2]_2$ and $7d'[5/2]_2$ levels. In the same manner, the $6\mathrm{p}[K]_J$ and $7\mathrm{P}[K]_J$ levels interact with the $4\mathrm{f}[K]_J$ and $5\mathrm{f}[K]_J$, respectively. On the other hand, the level $7p\ ^3D_1$ is perturbed by the level $5d\ ^1P_1$.

The oscillator strength of a transition between two energy levels in an atom is proportional to the square of the dipole matrix element that connects the two levels as well as the energy difference between these two levels. The dipole matrix element depends on the radial integral M_{if} which necessitates to be distinguished for different multipoles transitions and requires a gauges of the radiation fields. Consistently, Babushkin [57–59] and Coulomb [55,56] gauges are commonly used in relativistic electrodynamics. In the non-relativistic case, these two gauges correspond to the length and velocity gauges, respectively. Howbeit, many researchers use the gauge invariance of the predicted results in order to appraise the accuracy of the calculations of the oscillator strengths. However, the gauge invariance is itself necessary in non-relativistic calculation, but insufficient condition to draw conclusions about the physical convergent of the predicted results in relativistic framework. Based on the accurate position of our levels, the corresponding oscillator strengths f_{ij} and transition rates A_{ij} (Einstein A-coefficient in s^{-1}) were calculated in the length $f_{ij}(l)$, $A_{ij}(l)$ and velocity $f_{ij}(v)$, $A_{ij}(v)$ of the electric dipole operator forms in Babushkin and Coulomb gauges, respectively. The oscillator strengths and transition rates are calculated for transitions from the ground state $3p^6\ ^1S_0$ to $3p^5(^2p)$ 5d, $3p^5(^2p)$ 6d, $3p^5(^2p)$ 7s and $3p^5(^2p)$ 8s states. Since the transitions from the ground state to the levels 4f and 5f are not allowed either by parity or electric dipole selection rules, we calculated all the oscillator strengths and transition rates in emission for $3p^5(^2P)3d$-$3p^5(^2P)4f$, $3p^5(^2P)3d$-$3p^5(^2P)5f$, $3p^5(^2P)4d$-$3p^5(^2P)4f$, $3p^5(^2P)4d$-$3p^5(^2P)5f$, $3p^5(^2P)5d$-$3p^5(^2P)4f$, $3p^5(^2P)5d$-$3p^5(^2P)5f$, $3p^5(^2P)6d$-$3p^5(^2P)4f$ and $3p^5(^2P)6d$-$3p^5(^2P)5f$. Similarly, for the level 5p, we calculated all the transitions $3p^5(^2P)5d$-$3p^5(^2P)6p$, $3p^5(^2P)7s$-$3p^5(^2P)6p$, $3p^5(^2P)7s$-$3p^5(^2P)7p$ and $3p^5(^2P)8s$-$3p^5(^2P)7p$. This justifies the presence of the enormous number of transitions. The obtained results are displayed in table 2 of SM.

Clearly, for many levels, the agreement of the two gauges is satisfactory. However, the remaining disparity between length and velocity values of the oscillator strengths is expected in relativistic calculation, whereas the agreement is better in the non-relativistic calculations. This mismatch was also reported by Irimia and Froese Fischer [26] and Froese Fischer and Rubin [75]. This disparity may be ascribed to the fact that the amplitudes of the electric dipole transition in both Babushkin and Coulomb gauges are susceptible to diverse radial part of the wave functions. The radial dependence arises from the calculations of the transition matrix elements M_{if} of eq. (8) in Babushkin and Coulomb gauges. In Babushkin gauge, the wave functions in the larger r region participate considerably to the electric dipole transition amplitude. In Coulomb gauge, the transition matrix elements are supersensitive to the entire region of wave functions. In the interim, the large negligence in the integral of the transition matrix elements in Coulomb gauge significantly affects the velocity gauge results, on the one hand, and requires higher-quality wave functions, on the other hand. Under these circumstances, the transition rates in the Babushkin gauges are more reliable than those in the Coulomb gauge. Thence, the difference between the values of the oscillator strengths in the two gauges cannot be considered as an error bar by itself. In contrast, the difference between the Babushkin and Coulomb gauge values clarifies the fact that the core-polarization model is unsuitable to accurately compute the relative energy differences between the ground and the excited states as well as the departure and final levels when transitions are considered between two excited levels as interpreted by Irimia and Froese Fischer [26].

Conventionally, the error limit in the transition rates can be calculated by taking the entire difference between the Babushkin and Coulomb gauge final values. For strong transitions, the uncertainty is less than 2%, whereas for weak transitions it is of order 10%.

Over all, the two gauges display different energy dependence, the Coulomb gauge is strongly energy-dependent, while the Babushkin gauge is less dependent on the calculated values of the transition energy. For that reason, the Babushkin gauge is usually adopted by researchers in order to compare the theoretical values of the oscillator strengths and transition rates with the experimental ones, on the one hand, and in unsaturated calculations, on the other hand. Moreover, the calculations of energy level differences demand well-balanced orbital sets and typically necessitate highly extensive multiconfiguration expansions. If a common set of orbitals is used for both states, the results in both gauges converge and will be in fairly good agreement with the experimental ones.

The comparison of our predicted results of the oscillator strengths with the experimental and other theoretical results is essential to test the reliability and precision of our results. Table 3 of SM exhibits our theoretical results along with available data from other experiments and computations for the levels $7s[3/2]_1$, $7s'[1/2]_1$, $5d[1/2]_1$, $5d[3/2]_1$, $5d'[1/2]_1$ and $6d[3/2]_1$, respectively. Various experimental techniques have been used to measure the optical oscillator strength and different theoretical procedures have been used to calculate the oscillator strength in Coulomb and Babushkin gauge as one can see in table 3 of SM and the corresponding references therein. Unfortunately, these are the only experimental and theoretical data available for comparisons. In view of these comparisons, a clear disparity is observed between experimental-experimental data, theoretical-theoretical data as well as experimental-theoretical data. So that, one may raise the question whether the oscillator strengths were properly extracted in these experiments or accurately predicted in these theoretical works. This indicates that the available data are insufficient to judge the accuracy of our results and reveals the demand for more data in order to pass the judgment on the accuracy of all tabulated values.

The accuracy of the calculated transition rates is more difficult to estimate. The disparities between the Coulomb and Babushkin gauges indicate that the larger transitions rate causes the shorter lifetime data to be accurate. In general, theoretical lifetime data less than 30 ns usually agree with the experimental data [77]. Whilst, the theoretical values of the transition rates calculated in the Babushkin gauge are crowded in the direction of the external constitutions of the electronic wave functions, the Coulomb gauge values crowd additional internal constitutions of the wave functions, whereas the core-core effects take place. In order to check the accuracy of our results, we compare them with those measured experimentally or calculated theoretically by other researcher as shown in table 4 of SM. Our results for the lifetime of $3p^5\ nd$ ($n = 5, 6$) are compared with those measured experimentally by Das and Karmakar [41], and Zurro and co-workers [78] or calculated theoretically by Zurro and co-workers [78] and Gruzdev and Loginov [79]. Due to a shortage of both experimental and theoretical data in the case of argon, we did not find any more values of the lifetime for the other levels under consideration. We can see from this table that the agreement between our results and the theoretical ones of Gruzdev and Loginov is quite good whereas the agreement between our results and the experimental results is not as good as that with the theoretical results. The Zurro and co-workers [78] calculations were fulfilled in Coulomb gauge and Racah-coupling by ignoring the configuration interaction as well as the electron correlation. For that reason, our theoretical results disagree with their theoretical ones. The calculations of Gruzdev and Loginov [80] were performed by employing Hartree-Fock radial functions in order to predict the energy matrices for the electron-electron electrostatic interaction in addition to the spin-orbit interaction. The overlapping of configuration is taken into account in their treatment. The considerable disparities between the different forms of the dipole matrix elements arising in their calculation might be due to the insufficiency of the technique that they used for taking into account the overlap of configurations and should be removed in order to obtain better accuracy. Further investigation of the disparity between our theoretical results and the experimental tabulated values can be justified as follows: Firstly, the lifetime of our levels is calculated relative to the ground state $3p^6\ ^1S_0$, whereas the departure levels of the experimental values are unknown. Secondly, the experimental transition wavelengths to achieve the levels $5d[1/2]_1$ and $5d'[3/2]_1$ are 5607.7 and 5073.1 Å, respectively. These wavelengths correspond to wave numbers 17832.623 and 19711.81329 cm^{-1}, which do not match any transitions corresponding to the experimental measurements. Therefore, we may raise the question whether the lifetimes of these levels were adequately extracted in these experiments. Thus, more experimental results for the levels in argon are required in order to draw a judgement on the accuracy of our theoretical results. However, the quite good agreement between our results and the theoretical results of other researchers makes us confident in the fact that no unsuspected systematic errors could affect our own results.

After all, the atomic transition probabilities (Einstein coefficients) are unknown for many levels of the argon atom. However, their values are needed for spectroscopic diagnosis of argon plasmas and for calculating the plasma state, especially in cases where deviations from local thermodynamics equilibrium occur, which is generally the case in plasma physics laboratory.

5.3 Computation of the relativistic hyperfine interaction constants A_{MD} and B_{EQ} and Landé g_J-factor

For each level belonging to the series $n\mathrm{f}[K]_J$ $(n = 4, 5)$, $n\mathrm{d}[K]_J$ $(n = 5, 6)$, $n\mathrm{p}[K]_J$ $(n = 6, 7)$ and $n\mathrm{s}[K]_J$ $(n = 7, 8)$, we computed the A_{MD} and B_{EQ} hyperfine interaction constants in MHz and the Landé g_J-factor. The obtained results are displayed in table 5 of SM. In table 6 of SM, we display our theoretical results of the Relativistic Landé g-factor of $4\mathrm{f}[5/2]_3$, $^3\mathrm{D}_3$, $6\mathrm{p}[5/2]_2$ $^3\mathrm{D}_2$, $6\mathrm{p}[3/2]_1$ $^3\mathrm{D}_1$, $6\mathrm{d}[1/2]_1$ $^3\mathrm{P}_1$ and $6\mathrm{d}[3/2]_1$ $^3\mathrm{D}_1$ levels with other experimental [15] and NIST [38] results. However, we did not find any more values for our levels available in the literature. In sight of this comparison, one can see that our result for the $4\mathrm{f}[5/2]_3$ level is in good agreement with NIST [38]. While our result for the $6\mathrm{d}[1/2]_1$ $^3\mathrm{P}_1$ level is in good agreement with the experimental one [15], our result for the $6\mathrm{d}[3/2]_1$ $^3\mathrm{D}_1$ disagree with the corresponding experimental one. This disagreement due to the pure spin-orbit coupling between the level in question and the level $6\mathrm{d}'[3/2]_1$ $^1\mathrm{P}_1$. Moreover, the agreement between our results for the $6\mathrm{p}[5/2]_2$ $^3\mathrm{D}_2$ and $6\mathrm{p}[3/2]_1$ $^3\mathrm{D}_1$ levels and NIST ones is quite good. As on the one hand, this disparity might be due to the interaction between the $6\mathrm{p}[K]_J$ and $4\mathrm{f}[K]_J$ levels, on the other hand, if two Landé g-factors of a lower s-level and a higher p-level are not sufficiently different, the two corresponding values overlap due to the interaction with the close s- and p-configurations as well as the far ones. However, our values given in tables 5 and 6 of SM were calculated by assuming that the $3\mathrm{p}^5 6\mathrm{p}$ configuration is isolated from neighboring configurations. By the way, Anismova and Semenov [80] described a parametric approach for the levels of the $2\mathrm{p}^5 4\mathrm{p}$ configuration of the neon atom. In this approach, they included in addition to the spin-orbit interaction, the spin-spin, spin-other-orbit and orbit-orbit interactions. These interactions guided them to adjust twelve parameters from only ten energy levels. However, their treatment did not lead to a considerable amelioration of the general agreement between experimental as well as theoretical values of the Landé g_J-factor of neon. Therefore, we did not employ their treatment in the case of the argon atom.

For the levels under consideration, up to the present, only very few experimental and theoretical data of the Landé g-factor were available, whereas there are no other experimental or theoretical data of the magnetic dipole moment and the electric quadrupole hyperfine constants available for comparison. The scarceness of these data reveals the fact that our data for the levels under consideration are reported here for the first time. Therefore, we believe that our new, fairly complete, set of precise and reliable theoretical Landé g-factor and the hyperfine constants values should stimulate further theoretical and experimental works. Also, they could be used as a guide for experimentalists in identifying the fine-structure levels in their future work.

6 Conclusion

Based on the fully relativistic (four-component) multiconfiguration Dirac-Hartree-Fock (MCDHF) method, we calculated the energy of the levels belonging to the autoionizing intermediate Rydberg series $n\mathrm{f}[K]_J$ $(n = 4, 5)$, $n\mathrm{d}[K]_J$ $(n = 5, 6)$, $n\mathrm{p}[K]_J$ $(n = 6, 7)$ and $n\mathrm{s}[K]_J$ $(n = 7, 8)$ relative to the ground state $3\mathrm{p}^6$ $^1\mathrm{S}_0$ for the neutral argon atom. The correction of energy due to the dominant part of the radiative quantum electrodynamics effects included transverse photon Breit interaction and vacuum photon polarization self-energy; the core-polarization and electron correlation effects are also included. The accuracy of the obtained results is monitored by comparing them with those measured experimentally by Minnhagen [11], a reasonably satisfactory agreement is observed.

Based on the accuracy of the position of the energy levels, we calculated the oscillator strengths and transitions rate in both Coulomb (velocity) and Babushkin (length) gauges for many levels belonging to the configurations 4f, 5d, 6p, 7s, 5f, 6d, 7p and 8s. Since the transitions from the ground state to the levels 4f and 5f are not allowed either by parity or electric dipole selection rules, we calculated all the oscillator strengths and transitions rate in emission for $3\mathrm{p}^5(^2\mathrm{P})3\mathrm{d}$-$3\mathrm{p}^5(^2\mathrm{P})4\mathrm{f}$, $3\mathrm{p}^5(^2\mathrm{P})3\mathrm{d}$-$3\mathrm{p}^5(^2\mathrm{P})5\mathrm{f}$, $3\mathrm{p}^5(^2\mathrm{P})4\mathrm{d}$-$3\mathrm{p}^5(^2\mathrm{P})4\mathrm{f}$, $3\mathrm{p}^5(^2\mathrm{P})4\mathrm{d}$-$3\mathrm{p}^5(^2\mathrm{P})5\mathrm{f}$, $3\mathrm{p}^5(^2\mathrm{P})5\mathrm{d}$-$3\mathrm{p}^5(^2\mathrm{P})4\mathrm{f}$, $3\mathrm{p}^5(^2\mathrm{P})5\mathrm{d}$-$3\mathrm{p}^5(^2\mathrm{P})5\mathrm{f}$, $3\mathrm{p}^5(^2\mathrm{P})6\mathrm{d}$-$3\mathrm{p}^5(^2\mathrm{P})4\mathrm{f}$ and $3\mathrm{p}^5(^2\mathrm{P})6\mathrm{d}$-$3\mathrm{p}^5(^2\mathrm{P})5\mathrm{f}$. Similarly, for the level 5p, we calculated all the transitions $3\mathrm{p}^5(^2\mathrm{P})5\mathrm{d}$-$3\mathrm{p}^5(^2\mathrm{P})6\mathrm{p}$, $3\mathrm{p}^5(^2\mathrm{P})7\mathrm{s}$-$3\mathrm{p}^5(^2\mathrm{P})6\mathrm{p}$, $3\mathrm{p}^5(^2\mathrm{P})7\mathrm{s}$-$3\mathrm{p}^5(^2\mathrm{P})7\mathrm{p}$ and $3\mathrm{p}^5(^2\mathrm{P})8\mathrm{s}$-$3\mathrm{p}^5(^2\mathrm{P})7\mathrm{p}$.

Also, we computed the A_{MD} and B_{EQ} hyperfine interaction constants in MHz and the Landé g_J-factor for each level belonging to the series $n\mathrm{f}[K]_J$ $(n = 4, 5)$, $n\mathrm{d}[K]_J$ $(n = 5, 6)$, $n\mathrm{p}[K]_J$ $(n = 6, 7)$ and $n\mathrm{s}[K]_J$ $(n = 7, 8)$ relative to the ground state $3\mathrm{p}^6$ $^1\mathrm{S}_0$.

Insofar as experimental or theoretical values by other authors exist, comparison is made with those values; it turns out that for most of the levels, a rather satisfying agreement with these experimental and theoretical results is obtained, thus confirming the reliability of our data.

The atomic data presented in this paper with those published in our previous paper [28], constitute a complete set of atomic data of the neutral argon atom. This set provides a benchmark for both experimental measurements and theoretical calculations, as well as data for applications in plasma physics, atmospheric sciences, and astrophysics.

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